Woojeong Jin

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RESEARCH INTERESTS

My research focuses on **real-time voice agents**, **speech and audio models**, and **large language models**. I am particularly interested in integrating speech capabilities into large language models to build conversational AI systems that can listen, understand, and respond naturally in real-time. My work spans speech-to-speech models, automatic speech recognition (ASR), text-to-speech (TTS), and multimodal understanding, with a foundation in vision-language learning and low-resource generalization.

PROFESSIONAL EXPERIENCE

Senior Applied Scientist at **Hippocratic AI**, Palo Alto, California — Jun. 2024 – Present

- **Focus:** Real-time Voice Agents for Healthcare (ASR, Audio-LLM, Real-time Inference)
  - Trained and built contextual ASR system with decoder-only Audio-LLM integrating multi-turn dialogue context and personalized user biasing; achieved 50% WER reduction on medical domain and 85% average component error reduction on production calls.
  - Developed uncertainty-aware Single Word Correction module (phonetic confusion re-scoring, 92% error reduction) and targeted clarification system for residual ASR error recovery (88% reduction), forming layered safety defenses for clinical voice agents.
  - Trained end-to-end speech-to-response model (400B+ parameters) integrating speech understanding with LLM reasoning for low-latency voice agent interaction.

Research Intern at **Microsoft Research (MSR)**, Redmond, Washington — Jan. 2022 – Apr. 2022

- **Mentors:** Subho Mukherjee, Yu Cheng, Yelong Shen, and Ahmed H. Awadallah
- **Project:** Grounded Vision-language Pre-training via Aligning Text and Image Regions
  - Developed grounded VL pre-training method aligning text spans to image regions, enabling zero-shot transfer across 5 downstream tasks.
  - **Publication:** One preprint [P2] at arXiv.

Research Intern at **Microsoft Azure AI**, Redmond, Washington — Jun. 2021 – Jan. 2022

- **Mentors:** Yu Cheng, Yelong Shen, and Weizhu Chen
- **Project:** Low-resource Prompt-based Learning for Vision-Language Models
  - Proposed prompt-based method enabling 246× smaller models to match large VL model performance in zero-/few-shot settings.
  - **Publication:** One paper [C10] published at *ACL 2022*.

Research Intern at **Meta AI**, Menlo Park, California — May. 2020 – Aug. 2020

- **Mentors:** Hamed Firooz and Maziar Sanjabi
- **Project:** Knowledge Distillation for Multimodal Understanding
  - Investigated knowledge distillation for multimodal understanding through saliency-aware techniques, analyzing the importance and effects of each modality on distilled model performance.
  - **Publication:** One paper [C8] published at *EMNLP 2021 Findings* and One workshop paper [W2] published at *MAI@NAACL 2021*.

Research Assistant at **University of Southern California**, Los Angeles — Aug. 2018 – May. 2024

- **Advisor:** Xiang Ren
- Led research across multimodal few-shot learning [C10, P2], cross-modal knowledge transfer [C9], autoregressive temporal reasoning [C7, C6], and structured relational inference [C3–C5]; produced 10+ publications at ACL, EMNLP, and IJCAI.

EDUCATION

University of Southern California, Los Angeles, California — Aug. 2018 – May. 2024

- Ph.D. in Computer Science
- **Advisor:** Xiang Ren
- **Dissertation:** Bridging the Visual Reasoning Gaps in Multi-modal Models

Seoul National University, Seoul, Korea — Feb. 2017

- B.S. in Electrical and Computer Engineering

PUBLICATIONS

11 first-author publications. 12 conference, 4 journal, 4 workshop, 2 preprint, 1 tech report.

TECHNICAL REPORTS

[T1] Subhabrata Mukherjee, Markel Sanz Ausin, Kriti Aggarwal, Debajyoti Datta, Shanil Puri, Woojeong Jin, Tanmay Laud, Neha Manjunath, Jiayuan Ding, Bibek Paudel, Jan Schellenberger, Frazier Huo, Walter Shen, Nima Shirazian, Nate Potter, Sathvik Perkari, Darya Filippova, Anton Morozov, Austin Mease, Ghada Shakir, Alex Miller, Juliana Ghukasyan, Mariska Raglow-Defranco, Maggie Taylor, Herprit Mahal, Jonathan Agnew. Perfecting Human–AI Interaction at Clinical Scale: Turning Production Signals into Safer, More Human Conversations. January 2026. *Contextual ASR with Audio-LLM, single-word correction, and targeted clarification for voice agents deployed on 10M+ patient calls.*

PREPRINTS

[P2] Woojeong Jin, Subhabrata Mukherjee, Yu Cheng, Yelong Shen, Weizhu Chen, Ahmed Hassan Awadallah, Damien Jose, and Xiang Ren. GRILL: Grounded Vision-language Pre-training via Aligning Text and Image Regions. *arXiv preprint*.

[P1] Woojeong Jin, Tejas Srinivasan, Jesse Thomason, and Xiang Ren. WinoViz: Probing Visual Properties of Objects Under Different States. *arXiv preprint*.

REFEREED CONFERENCE PAPERS

[C12] Dong-Ho Lee, Kian Ahrabian, Woojeong Jin, Fred Morstatter, and Jay Pujara. Temporal Knowledge Graph Forecasting Without Knowledge Using In-Context Learning. *EMNLP 2023*.

[C11] Jihyung Moon*, Dong-Ho Lee*, Hyundong J. Cho, Woojeong Jin, Chan Young Park, Minwoo Kim, Jay Pujara and Sungjoon Park. Analyzing Norm Violations in Real-Time Live-Streaming Chat. *EMNLP 2023*.

[C10] Woojeong Jin, Yu Cheng, Yelong Shen, Weizhu Chen, and Xiang Ren. A Good Prompt Is Worth Millions of Parameters: Low-resource Prompt-based Learning for Vision-Language Models. *ACL 2022 (long)*.

[C9] Woojeong Jin*, Dong-Ho Lee*, Chenguang Zhu, Jay Pujara, and Xiang Ren. Leveraging Visual Knowledge in Language Tasks: An Empirical Study on Intermediate Pre-training for Cross-Modal Knowledge Transfer. *ACL 2022 (long)*.

[C8] Woojeong Jin, Maziar Sanjabi, Shaoliang Nie, Liang Tan, Xiang Ren, and Hamed Firooz. MSD: Saliency-aware Knowledge Distillation for Multimodal Understanding. *EMNLP 2021 Findings (long)*.

[C7] Woojeong Jin, Rahul Khanna, Suji Kim, Dong-Ho Lee, Fred Morstatter, Aram Galstyan, and Xiang Ren. ForecastQA: A Question Answering Challenge for Event Forecasting with Temporal Text Data. *ACL 2021 (long)*.

[C6] Woojeong Jin, Meng Qu, Xisen Jin, and Xiang Ren. Recurrent Event Network: Autoregressive Structure Inference over Temporal Knowledge Graphs. *EMNLP 2020 (long)*.

[C5] Sankalp Garg*, Navodita Sharma*, Woojeong Jin, and Xiang Ren. Temporal Attribute Prediction via Joint Modeling of Multi-Relational Structure Evolution. *IJCAI 2020*.

[C4] Cong Fu, Tong Chen, Meng Qu, Woojeong Jin, and Xiang Ren. Collaborative Policy Learning for Open Knowledge Graph Reasoning. *EMNLP 2019 (long)*.

[C3] Weizhi Ma, Min Zhang, Yue Cao, Woojeong Jin, Chenyang Wang, Yiqun Liu, Shaoping Ma, and Xiang Ren. Jointly Learning Explainable Rules for Recommendation with Knowledge Graph. *TheWebConf 2019*.

[C2] Minji Yoon, Woojeong Jin, and U Kang. Fast and Accurate Random Walk with Restart on Dynamic Graphs with Guarantees. *TheWebConf 2018*.

[C1] Jinhong Jung, Woojeong Jin, Lee Sael, and U Kang. Personalized Ranking in Signed Networks using Signed Random Walk with Restart. *ICDM 2016*.

REFEREED JOURNAL PAPERS

[J4] Daniel M Benjamin, Fred Morstatter, Ali E Abbas, Andres Abeliuk, Pavel Atanasov, Stephen Bennett, Andreas Beger, Saurabh Birari, David V Budescu, Michele Catasta, Emilio Ferrara, Lucas Haravitch, Mark Himmelstein, KSM Tozammel Hossain, Yuzhong Huang, Woojeong Jin, Regina Joseph, Jure Leskovec, Akira Matsui, Mehrnoosh Mirtaheri, Xiang Ren, Gleb Satyukov, Rajiv Sethi, Amandeep Singh, Rok Sosic, Mark Steyvers, Pedro A Szekely, Michael D Ward, Aram Galstyan. Hybrid forecasting of geopolitical events. *AI Magazine*, 2023.

[J3] Jinhong Jung, Woojeong Jin, Ha-myung Park, and U Kang. Accurate Relational Reasoning in Edge-labeled Graphs by Multi-labeled Random Walk with Restart. *World Wide Web Journal*, 2020.

[J2] Woojeong Jin, Jinhong Jung, and U Kang. Supervised and Extended Restart in Random Walks for Ranking and Link Prediction in Networks. *PLOS ONE*, 2019.

[J1] Jinhong Jung, Woojeong Jin, and U Kang. Random Walk Based Ranking in Signed Social Networks: Model and Algorithms. *KAIS*, 2019.

WORKSHOPS

[W4] Jihyung Moon*, Dong-Ho Lee*, Hyundong J. Cho, Woojeong Jin, Chan Young Park, Minwoo Kim, Jay Pujara and Sungjoon Park. Analyzing Norm Violations in Real-Time Live-Streaming Chat. *NLP+CSS@EMNLP 2022*.

[W3] Woojeong Jin, Yu Cheng, Yelong Shen, Weizhu Chen, and Xiang Ren. A Good Prompt Is Worth Millions of Parameters: Low-resource Prompt-based Learning for Vision-Language Models. *MML@ACL 2022*.

[W2] Woojeong Jin, Maziar Sanjabi, Shaoliang Nie, Liang Tan, Xiang Ren, and Hamed Firooz. Modality-specific Distillation. *MAI@NAACL 2021*.

[W1] Woojeong Jin, Changlin Zhang, and Xiang Ren. Recurrent Event Network for Reasoning over Temporal Knowledge Graphs. *ICLR-RLGM 2019*.

AWARDS & HONORS

- Kwanjeong Educational Foundation Scholarship — Aug. 2018 – 2023
- USC Annenberg Graduate Fellowship — Aug. 2018 – 2023
- Merit-based Scholarship, SNU — 2014, 2015, 2016

TECHNICAL SKILLS

- **Languages:** Python, C++, CUDA
- **ML Frameworks:** PyTorch, TensorFlow, Hugging Face libraries, NeMo
- **Infrastructure:** Distributed training (FSDP, DeepSpeed), Inference serving (vLLM, SGLang)
- **Tools:** Git, Weights & Biases, Jupyter/Colab, LaTeX

PROFESSIONAL SERVICE

- **Reviewer:** NeurIPS, EMNLP, KDD, WSDM, CoNLL, AKBC, CVPR Workshop

[*CV compiled on 2026-04-30*]